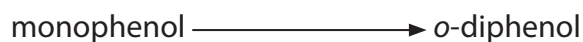


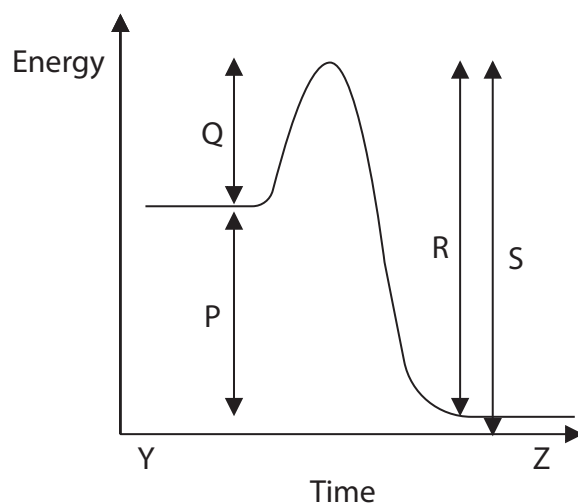
2 Polyphenol oxidase is an enzyme found in avocados.

This enzyme causes green avocados to go brown when they are cut open.

The equation shows the reaction catalysed by this enzyme.



- (a) The graph shows the energy changes during this reaction, when the enzyme is not present.



- (i) Which arrow shows the activation energy for this reaction?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

- (ii) Which row of the table shows the molecule present at time Y and time Z on this graph?

(1)

	Time Y		Time Z	
☐ A	product	monophenol	reactant	o-diphenol
☐ B	product	o-diphenol	reactant	monophenol
☐ C	reactant	monophenol	product	o-diphenol
☐ D	reactant	o-diphenol	product	monophenol

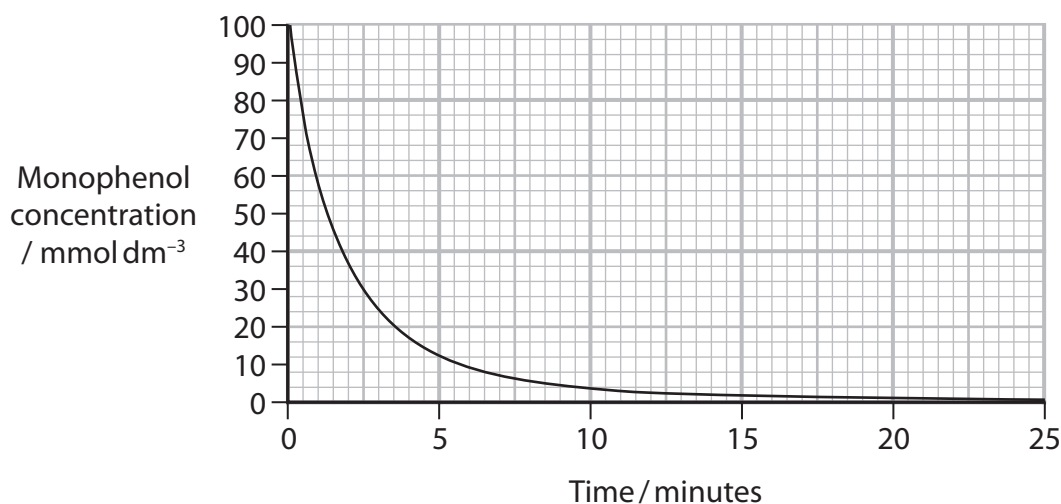


- (iii) Which row of the table shows how the length of the arrows P and Q on this graph would change when polyphenol oxidase is present?

(1)

Change in length of arrow		
	P	Q
☐ A	increase	decrease
☐ B	increase	no change
☐ C	no change	decrease
☐ D	no change	no change

- (b) The graph shows the change in monophenol concentration in the presence of polyphenol oxidase, at a temperature of 25 °C.



- (i) Calculate the rate of reaction at 3 minutes.

Draw a tangent to help with your calculation.

(2)

Answer mmol dm⁻³ min⁻¹

(ii) This reaction was carried out below the optimum temperature for the enzyme.

Explain what would happen to the shape of this curve if the temperature was increased to the optimum temperature for this enzyme.

(2)

(Total for Question 2 = 7 marks)

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